

CS2012 - Introduction to Object Oriented Programming

Form number	COURSE OUTLINE/ DOCUMENT	
QEC-ISOB-D1-CW 0.4		
COURSE INSTRUCTOR INFORMATION	Name	Ms. Magdalen Gloria Miranda
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DEGREE INFORMATION	Program	Batch	Section(s)			Semester	Fall
	BS Financial Technology	BSFT03	B			Year	2026

COURSE INFORMATION	Course Category C- Core/ E-Elective		Code	Title	Credit hours
	Not specified		CS2012	Introduction to Object Oriented Programming	3+1
	Prerequisite(s)		As per Academic policies		
	TA Required (Yes/ No)	No. of TA(s)	Brief Justification		
	Yes	1	To help in evaluating and analyzing different tasks		

TEXT BOOK(s) INFORMATION	Title of Book		Object-Oriented Programming in Python	Edition
	Author(s)		Michael H. Goldwasser & David Letscher	Latest
	Publisher		Not specified in source	
Reference Book(s)	1.	Title of Book	Python Crash Course — Eric Matthes	
		Imprint details	Reference book	
	2.	Title of Book	Data Structures and Algorithms in Python — Goodrich, Tamassia & Goldwasser	
		Imprint details	Reference book	
	3.	Title of Book	Clean Code — Robert C. Martin	
		Imprint details	Reference book	
	Support Material(s)	a.	Fluent Python — Luciano Ramalho	
		b.	Reference book	
		c.	Lecture slides	
		d.	Python practical exercises using VS Code/PyCharm	

Brief Description of Course: <i>(not more than 250 words)</i>	This course provides students with a foundational understanding of object-oriented programming in software development. It covers classes, objects, encapsulation, inheritance, polymorphism, abstraction, data structures, searching and sorting, algorithm analysis, software engineering processes, testing, debugging, exception handling and data persistence. Students develop hands-on Python programs and a Digital Wallet & Banking Platform project to solve real-world financial technology problems. Emphasis is placed on program specifications, correctness, efficiency, data abstraction and good programming practices.
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Learning Outcome (LO): <i>(Brief & unambiguous-with reference to course objectives i.e.at least 5 LOs</i> <i>a. describe broad aspects of behavior which incorporate a wide range of knowledge and skill</i> <i>b. accomplished over time in several learning experiences</i> <i>c. refer to <u>DEMONSTRATIONS OF PERFORMANCE</u></i>	
1.	CLO1: Explain and apply the fundamental concepts of object-oriented programming, including classes, objects, encapsulation, inheritance, polymorphism, and abstraction (Understand, Apply) – PLO1
2.	CLO2: Develop object-oriented programs using essential data structures, including arrays, linked lists, stacks, and queues, to solve basic computational and financial technology problems (Apply) – PLO2
3.	CLO3: Apply algorithmic problem-solving techniques, including searching, sorting, and basic algorithm analysis, to design efficient object-oriented solutions (Apply, Analyze) – PLO3
4.	CLO4: Implement, test, and debug modular object-oriented applications using good programming practices (Apply) – PLO3

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9
CLO 1	X								
CLO 2		X							
CLO 3			X						
CLO 4			X						

Courseware Structure: (Mark X where applies)

Lecture (X) | Multimedia (X) | Exercise (X) | IT Labs (X) | Case Studies (X) | Individual Assignment (X) | Group Presentation (X)

Hands-on Python programming, project development, testing/debugging, algorithm exercises and presentations.

Courseware Structure: (Mark X where applies)

Lecture (Lect)	Multimedia (MM)	Exercise (Exer)	IT Labs (Lab)	Case Studies (CAS)	Individual Assignment (Assign)	Group Presentation (G-Pres)	Any other Medium
X	X	X	X	X	X	X	Guest Lectures

COURSE CONTENTS:

Weeks	Contents/Topics	**Courseware Events (MM/ IT Lab/Case Study/ Assignment/ Presentation etc.)	Associated CLO(s)
Week 1	Introduction to Object-Oriented Programming • Procedural vs. OOP paradigms • Encapsulation, inheritance, polymorphism, abstraction • Python environment (VS Code/PyCharm) • Digital Wallet & Banking Platform project kickoff	Lecture, Multimedia, IT Lab	CLO1
Week 2	Classes & Objects • Classes, attributes, methods and __init__ • Objects and instance/class members • Base Account class	Lecture, IT Lab, Project	CLO1, CLO4
Week 3	Encapsulation & Data Abstraction • Access-control conventions • @property getters/setters • Interface vs. implementation • Secure balance and PIN handling	Lecture, IT Lab, Project	CLO1, CLO4
Week 4	Inheritance • Base/derived classes • Overriding and super() • Single and multilevel inheritance • Savings, Current and Wallet Account subclasses	Lecture, IT Lab, Project	CLO1, CLO4
Week 5	Polymorphism • Overriding and duck typing • Dynamic behavior • Common process_transaction() method	Lecture, IT Lab, Project	CLO1, CLO4

Week 6	Abstract Classes & Interfaces • Python abc module • Interface-like contracts • PaymentMethod implementations	Lecture, IT Lab, Project	CLO1, CLO4
Mid Sessional I	Mid Sessional I	Exam	All CLOs covered
Week 7	Program Specifications, Correctness & Efficiency • Docstrings and pre/post-conditions • Correctness and verification • Big O time and space complexity • Project documentation	Lecture, Exercise, IT Lab	CLO3, CLO4
Week 8	Object Collections (Lists & Custom Containers) • Python lists and list comprehensions • Collections of business objects • Multiple Account objects	Lecture, IT Lab, Project	CLO2
Week 9	Linked Lists (OOP Implementation) • Node-based class design • Insertion, deletion, traversal • Transaction history implementation	Lecture, IT Lab, Project	CLO2
Week 10	Stacks & Queues (OOP Implementation) • Class-based stack and queue • LIFO/FIFO applications • Rollback/undo and pending-payment queue • Quiz 01	Lecture, IT Lab, Quiz	CLO2
Week 11	Searching & Sorting Algorithms • Linear and binary search • Bubble, selection and insertion sort • Efficiency for financial data • Search/sort transaction features • Assignment 01	Lecture, IT Lab, Assignment	CLO3
Week 12	Software Engineering Processes • SDLC, requirements, design, implementation, maintenance • Modular design and separation of concerns • Refactoring wallet modules • Mid Sessional II	Lecture, Case Study, Exam	CLO4
Week 13	Testing & Debugging OOP Programs • Common OOP bugs • unittest/pytest basics • Debugging techniques • Unit tests for wallet functions	Lecture, IT Lab, Project	CLO4
Week 14	Exception Handling & Robust Programming • try/except/else/finally • Custom exceptions • Defensive programming and edge cases • Quiz 02 and Assignment 02	Lecture, IT Lab, Quiz, Assignment	CLO4
Week 15	File Handling & Data Persistence • Text, CSV and JSON • Object serialization/deserialization • Persistence patterns • Save/reload accounts and transactions	Lecture, IT Lab, Project	CLO2, CLO4
Week 16	Course Review & Final Presentations • Review • Final project presentations • FINAL-TERM EXAM	Review, Group Presentation, Exam	All CLOs covered

****Courseware Events** column is subject to variations / appropriations

INDIGENOUS MATERIAL USED:

S. No	Material /Activity	Associated Topic
1	Python OOP Practical Exercises	Weeks 1–6: Core OOP
2	Digital Wallet & Banking Platform Project	Weeks 1–6: OOP design
3	Data Structure Implementation Templates	Weeks 8–10: Lists, Linked Lists, Stacks, Queues
4	Algorithm Exercise Sets	Week 11: Searching and Sorting
5	Software Design / Refactoring Templates	Week 12: SDLC and modular design
6	Unit Testing & Debugging Exercises	Week 13
7	Custom Exception Examples	Week 14
8	JSON/CSV Persistence Templates	Week 15

AI INTEGRATION AND TRANSPARENCY

S. No.	Course Component / Activity	AI Use by Students	AI Use by Instructor	Purpose of AI Use	AI Contribution (%)	Human Contribution (%)	Verification / Transparency
1	Lecture Preparation	N/A	Yes	Content development, examples, visual aids	50%	50%	Instructor revises AI content
2	Lecture Delivery	N/A	Optional	Interactive demonstrations, real-time examples	50%	50%	Instructor-led discussion and contextualization
3	Assignments	As per instructions for each assignment	Yes	Idea generation, brainstorming, feedback	Up to 20%	Minimum 80%	Students submit AI usage declaration
4	Case Analysis	As per instructions	Yes	Idea generation, brainstorming, feedback	Up to 20%	Minimum 80%	Students submit AI usage declaration
5	Project Report	As per instructions	Optional	Idea generation, brainstorming, feedback	Up to 10%	Minimum 90%	Students submit AI usage declaration
6	Presentations	N/A	N/A	N/A	N/A	N/A	N/A
7	Research Activities	N/A	N/A	N/A	N/A	N/A	N/A
8	Assessment Design	N/A	N/A	N/A	N/A	N/A	N/A
9	Feedback to Students	N/A	N/A	N/A	N/A	N/A	N/A
10	Administrative Tasks	N/A	N/A	N/A	N/A	N/A	N/A

AI TRANSPARENCY DECLARATION

Requirement	Yes / No
Students are informed about permitted AI uses in the course	Yes (As per instructor permission)
Students are required to disclose AI usage in assignments/projects	Yes (As per instructor permission)
AI-generated content is subject to verification and critical evaluation	Yes
AI is used to augment learning rather than replace learning outcomes	Yes

Instructor reviews and validates all AI-assisted teaching materials	Yes
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Recommended Web links:
Python Documentation
VS Code
PyCharm
unittest / pytest Documentation

Grading Criteria			
	ABSOLUTE Grading		RELATIVE Grading

Marks Distribution:

Particulars	% Marks	*Weight Ranges
1. Assignments + Quizzes		12%
2. First Sessional Exam		15%
3. Second Sessional Exam		15%
4. Project		8%
5. Final Exam		50%
6. Class Participation		
7. Total	100%	100%

Planned Courseware Events:

Particulars	Planned (Qz/As/Labs)	Remarks
1. Quizzes	2 (n-1 if required)	Quiz 01 and Quiz 02
2. Assignments	2 (n-1 if required)	Assignment 01 and Assignment 02
3. Presentations / Project	1	

**WeightRanges as defined in FLEX*

QUALIFYING ATTENDANCE	You must attend every class for your own personal benefit. Please refer to university policy of minimum attendance requirement. Failing to conform qualifying attendance threshold, the student will stand debarred from sitting in the examination and assigned with "F" Grade.
Academic and Moral Integrity:	All assignments should be your own work (or your group's when approved). PLAGIARISM will be awarded with "F" grade and/or reported to the University for academic and moral misconduct. To protect yourself, ALWAYS PROVIDE REFERENCES!

Instructions / Suggestions for STUDENTS for satisfactory progress in this course:

- ✓ FAST has a 100 percent attendance policy. However, students are allowed to sit in the final exam if they have at least 80 percent attendance.
- ✓ Do not schedule other appointments during your lecture time, still If you miss a lecture; Ask for a copy of the lecture notes. Get a good, readable and reliable copy and work though it carefully and diligently.
- ✓ Plagiarism in any part of the course can lead to an **F** in the whole course.
- ✓ On average, most students should find at least three hours outside of class for each class hour for satisfactory learning.
- ✓ **Note:** Counseling hours are displayed outside the doors of permanent faculty members for providing extra guidance to students. However, students are welcome to get help from the instructor at all times.
- ✓ If you encounter academic difficulties in your course, do not wait, contact your instructor or advisor in designated counseling hours and seek appropriate levels of assistance to help resolve your academic apprehensions.
- ✓ Inform your course instructor, advisor, administrator or staff person of your concerns and provide them with all relevant information on the challenges you are facing.
- ✓ Read the objectives and learning outcomes well; students are expected to attain the intended learning outcomes.
- ✓ Plan all of your activities; use the calendars, things to do list, reminders or other resources to help you keep track of important tasks, course assignments, quizzes, examination dates and deadline dates as well as your co-curricular activities.
- ✓ Students unable to submit an assignment by the due date should discuss the matter in advance with the course instructor. At the instructor's discretion, late assignments may not be accepted. Where late assignments are accepted, a late penalty of 10% per day will be levied.
- ✓ **Assignments/ Activities:** They are not meant simply for grades, but to reinforce your learning. Assignments are due on time. Each day late will lower your assignment grade by 10%. Apart from value of content, spelling, grammar, punctuation, and good presentation (printing and paper quality) will figure into your assignment grade.
- ✓ Copying other students' work or allowing your own work to be copied, both are considered cheating.
- ✓ You must logged in to your SLATE account at least once in every 24 hours for announcements, assignments, lecture notes and messages
- ✓ Check FLEX regularly for your attendance, and evaluations.
- ✓ You need to enter the lecture room fully awake and able to devote your best concentration, for this go to bed at reasonable times and take optimum sleep.
- ✓ Come to each lecture properly prepared, bring in last lectures' notes, text books, ball points/pencils, laptops and calculators etc
- ✓ Use the few minutes you usually have before the start of each class to review the prior lectures' notes and homework. This will save us valuable in-class time to work on new material.
- ✓ Chapters should be read and homework/exercises should be attempted before class.
- ✓ Do not get behind. You are encouraged to work with other students.
- ✓ The homework assigned is a minimum. You should always work extra hours on your own.
- ✓ Develop a learning habit rather than memorizing; work in groups, whenever appropriate.
- ✓ Apply the learned principles and gained knowledge; be creative in thinking.
- ✓ To guard against errors, please keep copies of the papers you turn in and retain all graded assignments for your reference.
- ✓ It is the responsibility and duty of students to become acquainted and abide by all the policies and rules of the FSM.

SPECIAL INSTRUCTIONS
(Specific to the subject
being taught)